

Extra 01

Exam Grade Table

Purpose

Summary table of exam scores, final grades, and average extensions broken down by GPA range and number of extensions used.

Output Description

Tabular output — printed to console and exported as a CSV.

Rows grouped by: GPA Range × Extensions Used

Columns: Avg Exam Score | Avg Final Grade | Student Count (n)

CSV output: GPA_Extension_Analysis.csv

Raw Values Summary

1. A Range (3.5 - 4.0) who used 0 extensions had an average exam score of 104.74 and an average final grade of 102.15 (count: 15)
2. A Range (3.5 - 4.0) who used 1 extension had an average exam score of 100.66 and an average final grade of 100.64 (count: 17)
3. A Range (3.5 - 4.0) who used 2 extensions had an average exam score of 99.11 and an average final grade of 101.02 (count: 17)
4. B Range (3.0 - 3.49) who used 0 extensions had an average exam score of 117 and an average final grade of 103.27 (count: 1)
5. B Range (3.0 - 3.49) who used 1 extension had an average exam score of 88.57 and an average final grade of 96.6 (count: 4)
6. B Range (3.0 - 3.49) who used 2 extensions had an average exam score of 89.15 and an average final grade of 92.48 (count: 12)
7. C/D Range (< 3.0) who used 0 extensions had an average exam score of 82.69 and an average final grade of 84.59 (count: 1)
8. C/D Range (< 3.0) who used 1 extension had an average exam score of 92 and an average final grade of 86.94 (count: 1)
9. C/D Range (< 3.0) who used 2 extensions had an average exam score of 75.36 and an average final grade of 88.25 (count: 9)

Extra 02

Exam Scores vs Extensions Used

Purpose

Does the number of extensions used correlate with final exam performance?

Graph

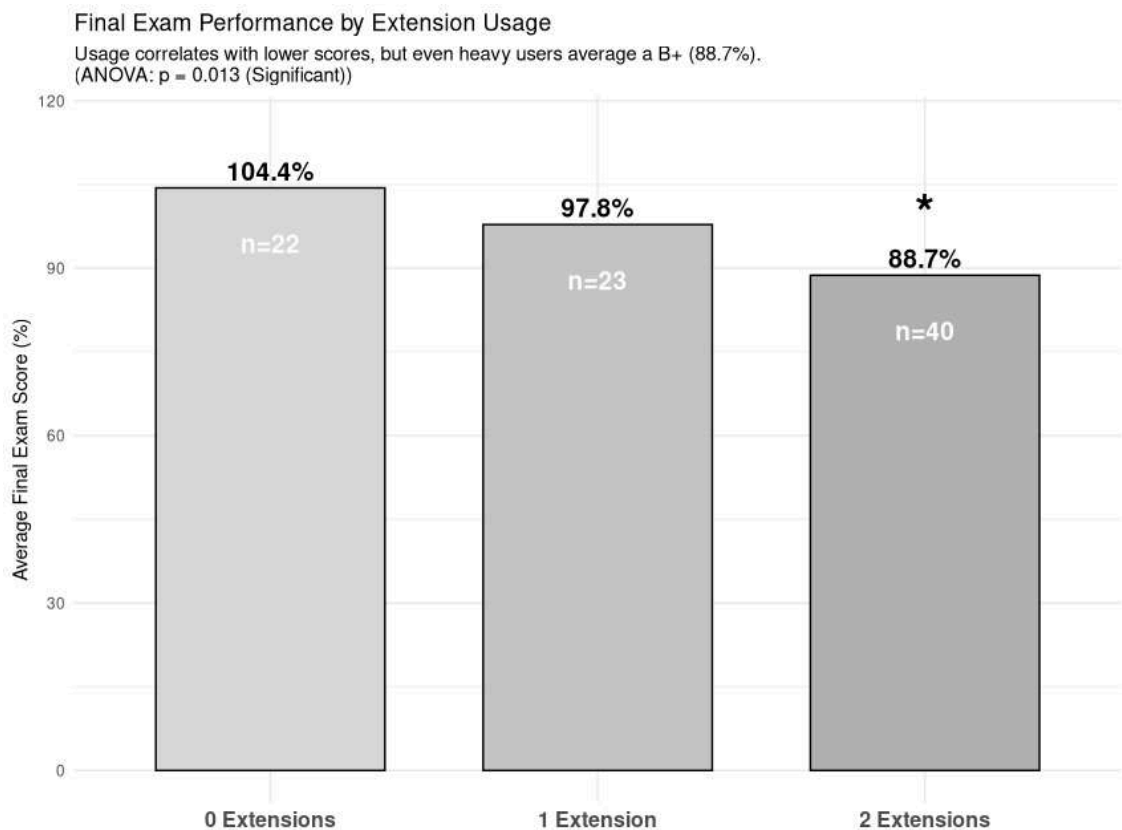


Figure 1: Average final exam score by number of extensions used.

Graph Description

Bar chart — Average final exam score per extension usage group.

X-axis: Extension group (0 / 1 / 2 Extensions)

Y-axis: Average final exam score (%)

n counts shown inside bars; mean % shown above bars.

Significance star (*) annotated on the graph.

Raw Values Summary

1. Those who took no extensions (count: 22) got a mean exam score of 104.4
2. Those who took 1 extensions (count: 23) got a mean exam score of 97.8
3. Those who took 2 extensions (count: 40) got a mean exam score of 88.7

Statistical Note

Test used: One-way ANOVA

Result:

$$p < 0.05$$

Extra 03

Exam Scores Controlled for GPA

Purpose

Does extension usage predict lower exam scores even after controlling for prior GPA?

Graph

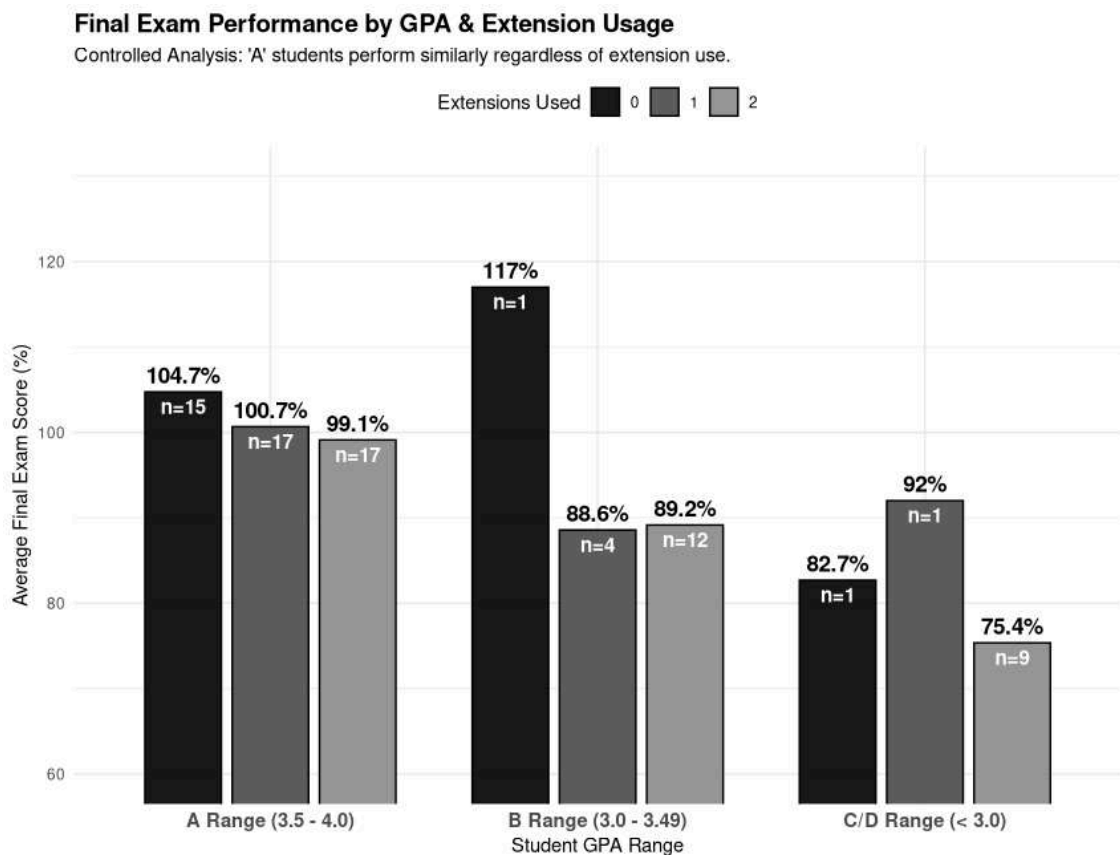


Figure 1: Average final exam score by GPA range and extension usage.

Graph Description

Grouped bar chart — Average final exam score by GPA range, grouped by extensions used.

X-axis: GPA range (A / B / C–D)

Y-axis: Average final exam score (%) — range 60–130

Grouped bars per GPA range, colored by extension count (0 / 1 / 2).

Score % labels above each bar; n= labels inside each bar.

Raw Values Summary

1. A Range (3.5 - 4.0) who used 0 extensions (count: 15) got a mean exam score of 104.74
2. A Range (3.5 - 4.0) who used 1 extension (count: 17) got a mean exam score of 100.66
3. A Range (3.5 - 4.0) who used 2 extensions (count: 17) got a mean exam score of 99.11
4. B Range (3.0 - 3.49) who used 0 extensions (count: 1) got a mean exam score of 117
5. B Range (3.0 - 3.49) who used 1 extension (count: 4) got a mean exam score of 88.57
6. B Range (3.0 - 3.49) who used 2 extensions (count: 12) got a mean exam score of 89.15
7. C/D Range (< 3.0) who used 0 extensions (count: 1) got a mean exam score of 82.69
8. C/D Range (< 3.0) who used 1 extension (count: 1) got a mean exam score of 92
9. C/D Range (< 3.0) who used 2 extensions (count: 9) got a mean exam score of 75.36

Research Question 1

GPA & Extension Requests

Question

Will students with lower starting GPAs more likely request extensions than students with higher starting GPAs?

Hypothesis

Students with lower starting GPAs are more likely to request extensions than students with higher starting GPAs.

Graph

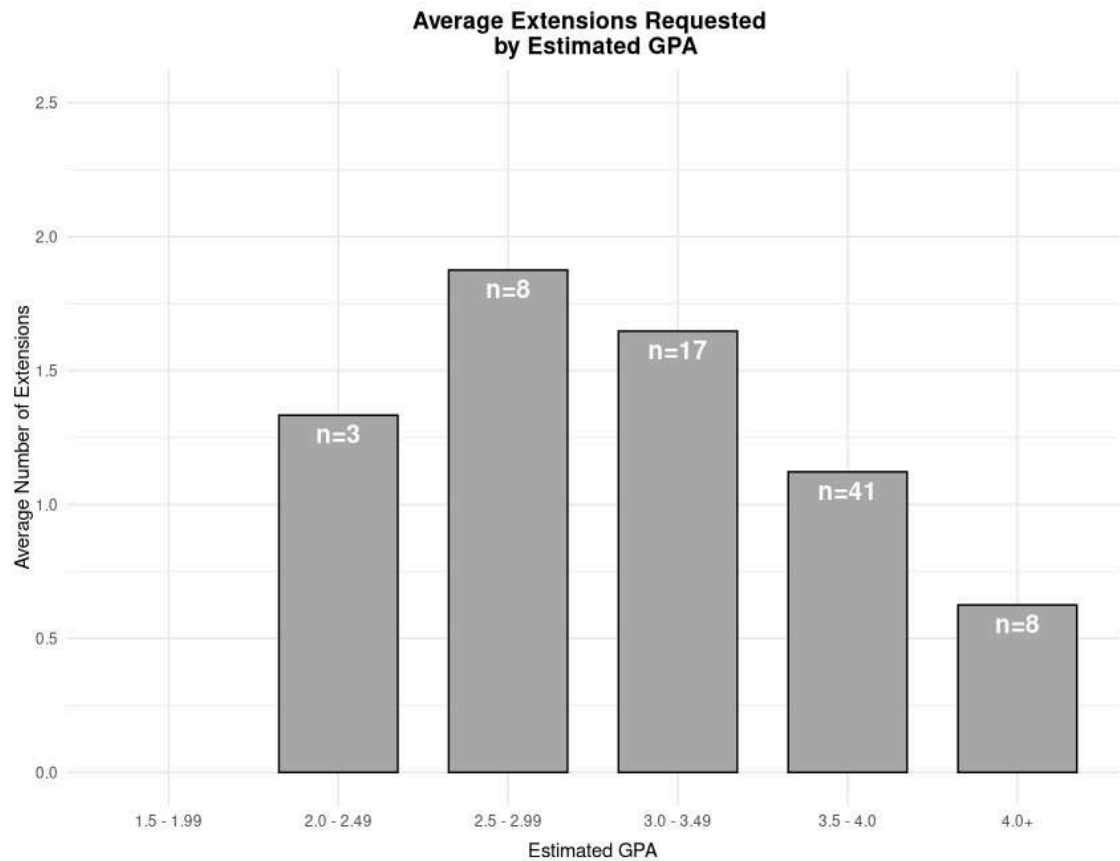


Figure 1: Average extensions requested by estimated GPA group.

Graph Description

Bar chart — Average number of extensions requested per GPA group.

X-axis: Estimated GPA range

Y-axis: Average number of extensions
n counts displayed inside each bar.

Raw Values Summary

1. GPA Group 2.0 – 2.49 (with a count of 3) has 1.3333 average extensions used
2. GPA Group 2.5 – 2.99 (with a count of 8) has 1.875 average extensions used
3. GPA Group 3.0 – 3.49 (with a count of 17) has 1.6471 average extensions used
4. GPA Group 3.5 – 4.0 (with a count of 41) has 1.122 average extensions used
5. GPA Group 4.0 + (with a count of 8) has 0.625 average extensions used

Statistical Note

Test used: Spearman Correlation

Result:

$$\rho = -0.43, p < 0.0001$$

Conclusion

Using a Spearman Correlation with a rho value of -0.43 and a P-value of < 0.0001 . This means there is a moderate negative correlation; as the GPA increases, the number of extensions requested decreases. The bar graph shows a clear downward trend. The reason there are no entries for the “1.5 - 1.99” rank is due to no IRB approved students answering this.

Research Question 2

Course Level & Extension Requests

Question

Will students in lower-division (freshman/sophomore) courses request extensions more frequently than those in upper-division (junior) courses?

Hypothesis

Students in lower-division (freshman/sophomore) courses will request extensions more frequently than those in upper-division (junior) courses.

Graph

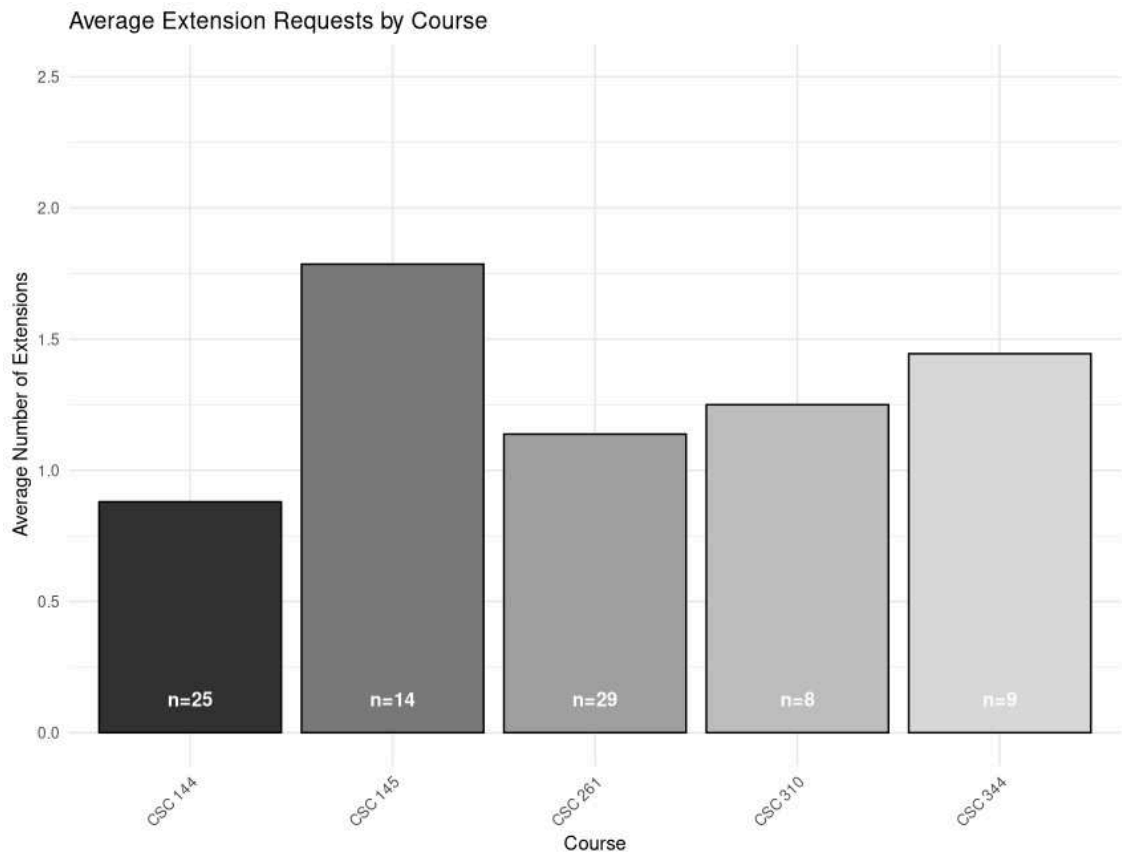


Figure 1: Average extension requests by course.

Graph Description

Bar chart — Average number of extensions requested per course.

X-axis: Course code

Y-axis: Average number of extensions

Error bars shown. n counts displayed at the bottom of each bar.

Raw Values Summary

1. Course CSC-144 (count: 25) had a mean of 0.88 extensions used
2. Course CSC-145 (count: 14) had a mean of 1.7857 extensions used
3. Course CSC-261 (count: 29) had a mean of 1.1379 extensions used
4. Course CSC-310 (count: 8) had a mean of 1.25 extensions used
5. Course CSC-344 (count: 9) had a mean of 1.4444 extensions used

Statistical Note

Test used: One-way ANOVA

Result:

$$F(4, 80) = 3.20, p = 0.017$$

Conclusion

Based on our tests, there is no significant correlation between extension usage and the courses, either upper or lower level. In CSC 145, there is a higher amount of extensions used, however, this could be due to the amount of assignments in the class. I was not satisfied with this conclusion, so I decided to perform another correlation between the grade level and the number of extensions used (see Research Question 9)

AI: “A one-way ANOVA was conducted to determine if the number of extension requests differed significantly across the different courses. The results indicated a marginally significant difference between the courses ($F(4, 80) = 3.20, p = 0.017$). While the p-value sits just above the strict 0.05 threshold for statistical significance, the data suggests a strong trending variance in extension usage depending on the specific class a student is taking.

Research Question 3

Extension Reasons

Question

What will be the most common reason students will request extensions?

Hypothesis

The most common reason students request extensions will be workload conflicts (e.g., multiple assignments due simultaneously).

Graph

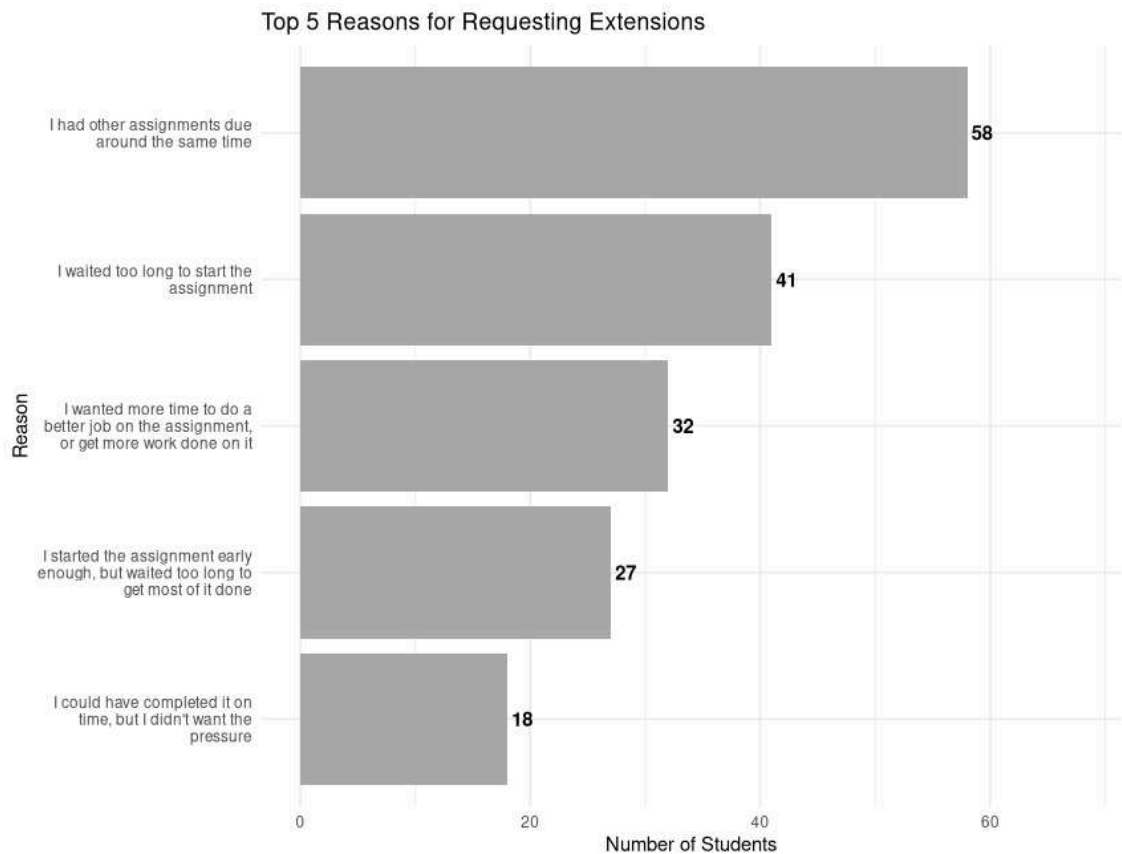


Figure 1: Top 5 reasons for requesting extensions.

Graph Description

Horizontal bar chart — Top 5 reasons for requesting extensions.

X-axis (flipped): Number of students

Y-axis (flipped): Reason category

Count labels displayed at the end of each bar.

Note: students may contribute to multiple reasons.

Raw Values Summary

1. "I had other assignments due around the same time" : 58 responses
2. "I waited too long to start the assignment" : 41 responses
3. "I wanted more time to do a better job on the assignment or get more work done on it" : 32 responses
4. "I started the assignment early enough, but waited too long to get most of it done" : 27 responses
5. "I could have completed it on time, but I didn't want the pressure" : 18 responses

Statistical Note

Test used: Frequency count / descriptive

Result: "I had other assignments due around the same time" : 78 responses

Conclusion

Yes, this hypothesis is supported. 78 students answered that they "had other assignments due around the same time" which was the top reason. Note, students could answer multiple reasons.

Research Question 4

Fairness & Psychological Safety

Question

Will the availability of automated, no-questions-asked extensions increase students' overall sense of fairness and psychological safety in the course?

Hypothesis

The availability of automated, no-questions-asked extensions will increase students' overall sense of fairness and psychological safety in the course.

Graph

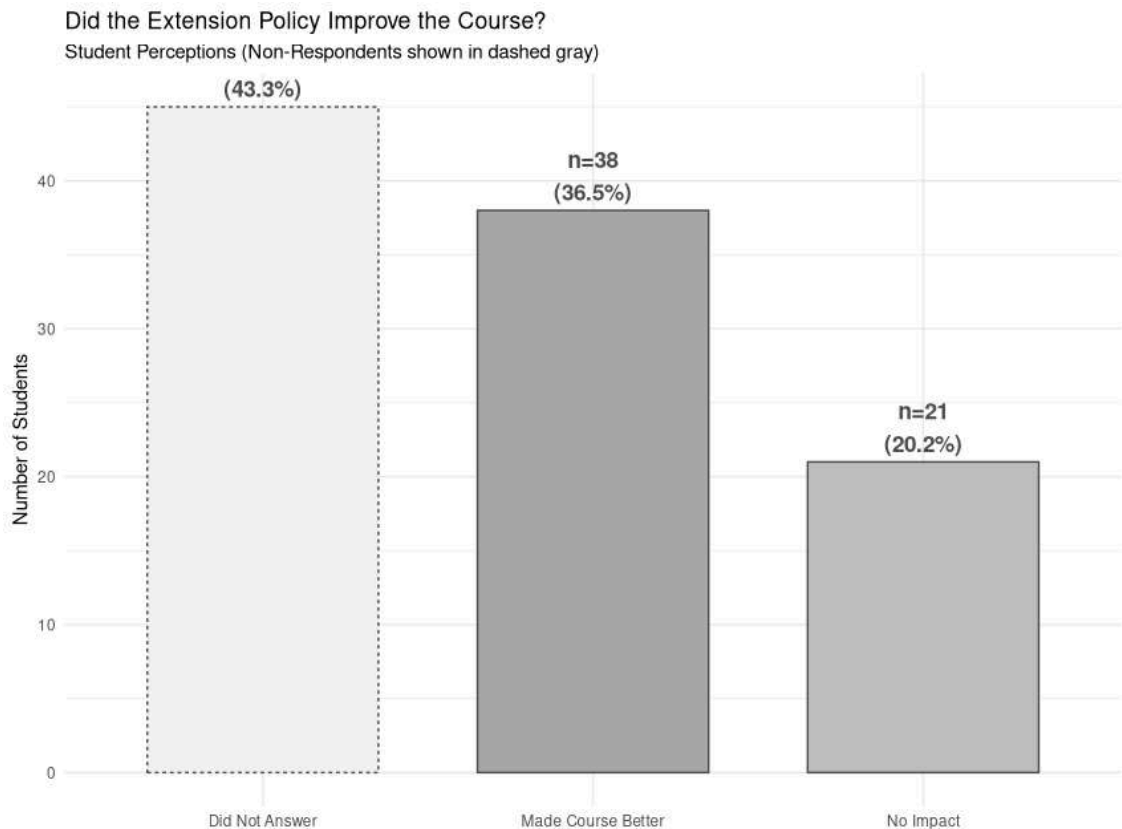


Figure 1: Student perceptions of the extension policy's impact on the course.

Graph Description

Bar chart — Student opinions on whether the extension policy improved the course.

X-axis: Opinion category (Made Course Better / No Impact / Made Course Worse / Did Not Answer)

Y-axis: Number of students

Labels show n= count and percentage for each bar.

Raw Values Summary

1. 43.3% (n = 45) of respondents didn't answer (oddly)
2. 36.5% (n=38) of respondents stated it made the course better
3. 20.2% (n=21) stated the policy had no impact on course quality

Statistical Note

Test used: Descriptive / frequency count / proportions test

Result:

$$p < 0.05$$

Conclusion

AI: “A one-sample proportion test resulted in a p-value of 0.0046. Because this p-value is well below the standard significance threshold of 0.05, we reject the null hypothesis that student opinions are evenly divided. This statistically significant result indicates that the overwhelming positive feedback is not due to random chance, allowing us to confidently conclude that the automated extension policy significantly increases students' overall sense of fairness and psychological safety in the course.”

Research Question 5

Assignment Value Perception

Question

Will students who use extensions perceive assignments as less valuable to their education than students who do not?

Hypothesis

Students who use extensions will **not** perceive assignments as less valuable to their education than students who do not.

Graph

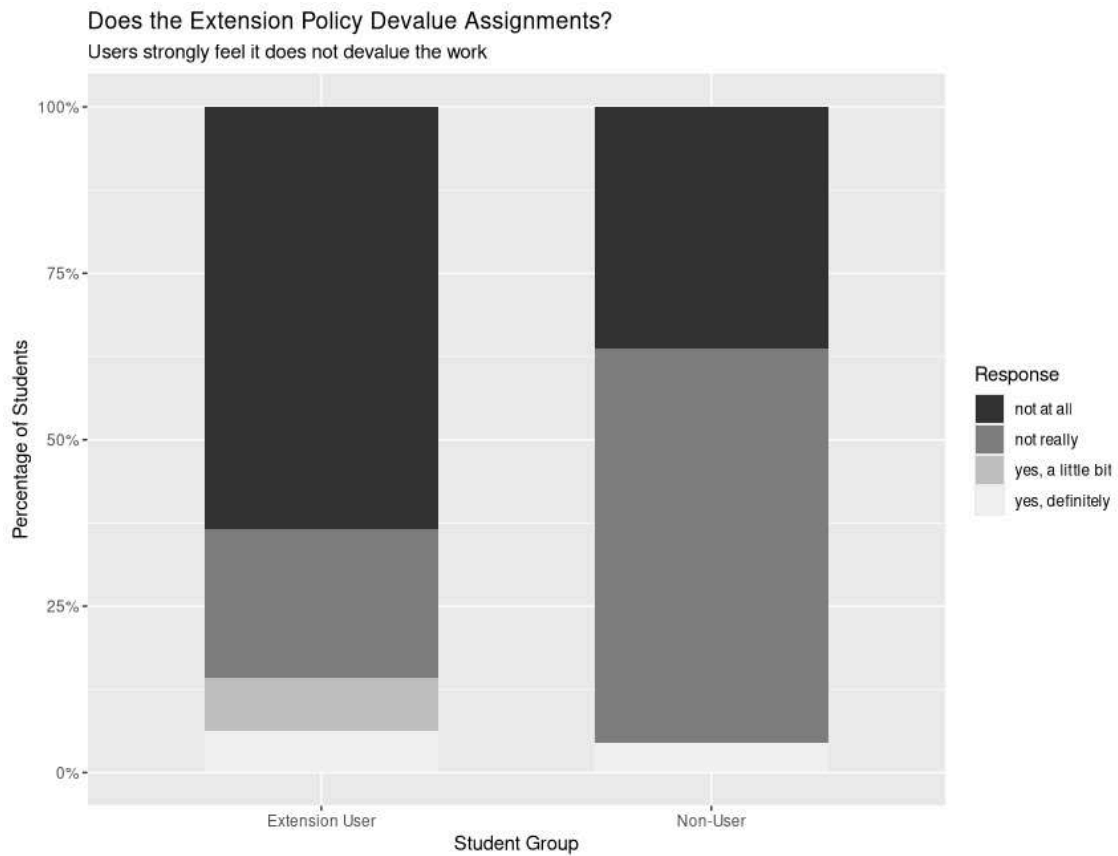


Figure 1: Perceived homework importance by extension user group.

Graph Description

Stacked proportional bar chart — Perceived homework importance by student group.

X-axis: Student group (Extension User / Non-User)

Y-axis: Percentage of students

Fill: Response level (not at all / not really / yes, a little bit / yes, definitely)

Raw Values Summary

Group sizes: Extension User $n = 63$, Non-User $n = 22$.

Extension User ($n = 63$):

1. “not at all” — 40 students (63.5%)
2. “not really” — 14 students (22.2%)
3. “yes, a little bit” — 5 students (7.9%)
4. “yes, definitely” — 4 students (6.3%)

Non-User ($n = 23$):

1. “not at all” — 8 students (36.4%)
2. “not really” — 13 students (59.1%)
3. “yes, definitely” — 1 student (4.5%)

Statistical Note

Test used: Mann-Whitney U (Wilcoxon)

Result:

$$p = 0.1174$$

Conclusion

A Mann-Whitney U test was conducted to determine if utilizing the extension policy led students to devalue the importance of their assignments. The analysis revealed no statistically significant difference in perceived assignment value between students who used extensions and those who did not ($p > 0.05$). Despite this, the data shows a shared consensus that the vast majority of students think that the policy’s flexibility does not directly impact their appreciation of the coursework.

Research Question 6

Submission Completion Rate

Question

Of the students who requested extensions, what percentage were never submitted versus the percentage that were submitted?

Graph

Do Extensions Lead to 'Ghosting'?

Completion Rate: 94.2 % - Students almost always do the work.

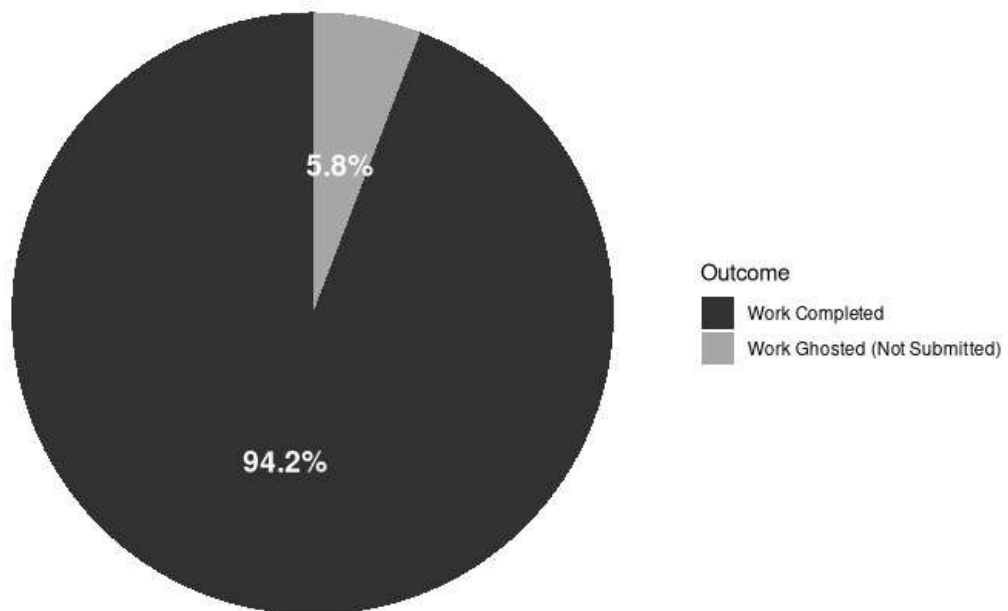


Figure 1: Submission outcomes for extension requests.

Graph Description

Pie chart — Submission outcomes for extension requests.

Slices: Work Completed vs Work Ghosted (Not Submitted)

Labels show percentage of total extension requests.

Raw Values Summary

1. Out of those who took extensions 94.2% (Count: 97) completed their extended work.
2. Out of those who took extensions 5.8% (Count: 6) did not complete their work (ghosted their work).

Conclusion

Students who requested an extension almost always did the work for the assignment in which the extension was used for. According to the percentages, 93.5% completed their work for the assignment an extension was used. 6.5% ghosted their assignment even after requesting an extension

Research Question 7

Protocol Compliance

Question

How many extension requests were submitted correctly and on time according to the protocol?

Graph

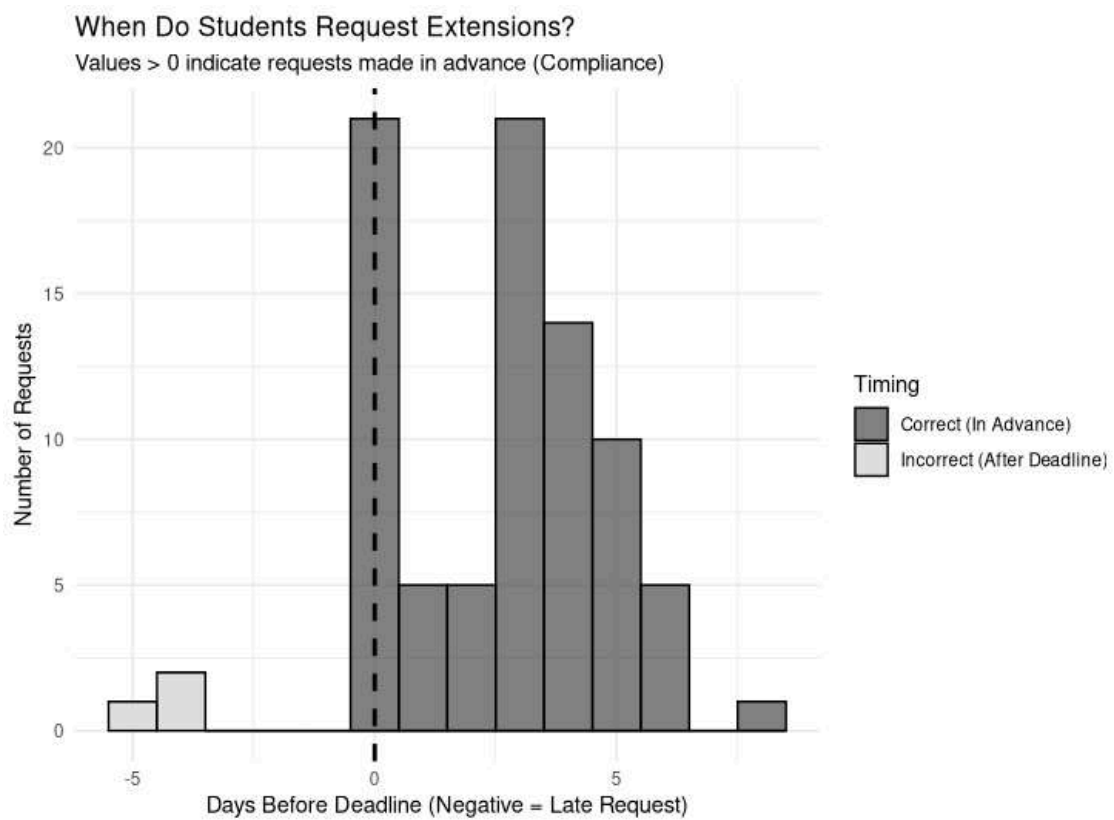


Figure 1: Timing of extension requests relative to the deadline.

Graph Description

Histogram — Timing of extension requests relative to deadline.

X-axis: Days before deadline (negative = late / after deadline)

Y-axis: Number of requests

Bars colored by compliance (Correct = In Advance / Incorrect = After Deadline).

Vertical dashed line at $x = 0$. Bin width: 1 day.

Raw Values Summary

Of the 85 students with recorded extension request timing, 82 (96.4%) submitted correctly in advance and 3 (3.5%) submitted after the deadline.

Among the 82 correct requests: 21 submitted on the day of the deadline (0 days before), 5 submitted 1 day before, 5 submitted 2 days before, 21 submitted 3 days before, 14 submitted 4 days before, 10 submitted 5 days before, 5 submitted 6 days before, and 1 submitted 8 days before.

Among the 3 incorrect requests: 2 were submitted 4 days after the deadline and 1 was submitted 5 days after the deadline.

The mean request timing across all 85 students was 2.46 days before the deadline.

Conclusion

Almost all of the extension requests were submitted correctly and **on time** before or right on the deadline. According to the graph, the students on the right of '0' correctly submitted the extension x amount of days before the deadline. The students on the left of '0' are the ones that submitted it incorrectly. There are significantly more blue bars which represent the amount of students who submitted correctly. This displays that our extension policy does not encourage laziness in class, but provides a way for students to excel while giving them more freedom in the course load.

Research Question 8

Year Level & Extension Usage

Question

Is there a correlation between grade level and the amount of extensions used?

Graph

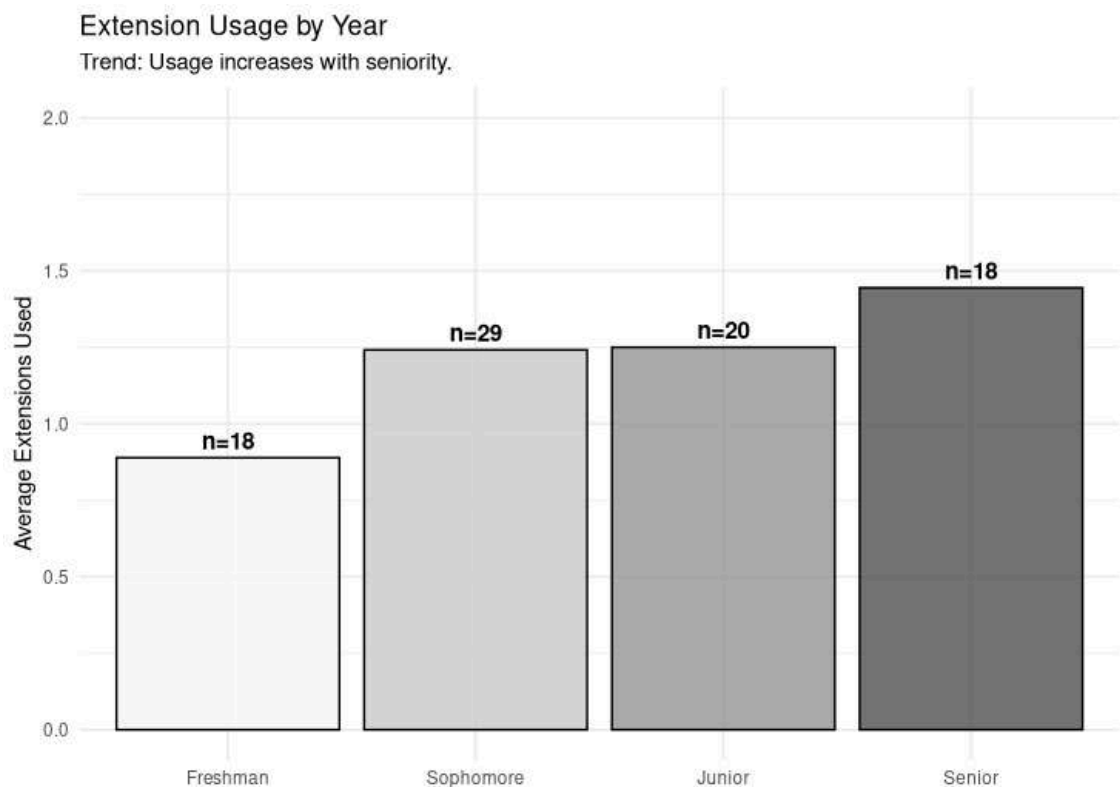


Figure 1: Average extension usage by academic year level.

Graph Description

Bar chart — Average extensions used per academic year level.

X-axis: Year level (Freshman / Sophomore / Junior / Senior)

Y-axis: Average extensions used

n counts displayed above each bar.

Raw Values Summary

1. The Freshmen (Count: 18) had an mean of 0.89 extensions used.
2. The Sophomores (Count: 29) had an mean of 1.24 extensions used.
3. The Juniors (Count: 20) had an mean of 1.25 extensions used.
4. The Seniors (Count: 18) had an mean of 1.44 extensions used.

Statistical Note

Test used: One-way ANOVA

Result:

$$p = 0.243$$

Conclusion

Though the data is not significant ($p > 0.05$), this still tells us something. Our data shows a positive correlation: as grade level increases, the amount of extensions used increases slightly. Our extensions policy is helpful for all students and is not necessarily a “safety net” or a “holding hand” for lowerclassmen students who just began college. Despite this, due to the insignificance proven, our extension policy can help all students alike without hindering performance or drastically changing performance.

Research Question 9

Achievement Gap

Question

Does the policy help close the gap between students with lower prior GPAs and high achievers?

Graph 1 — Extensions by GPA

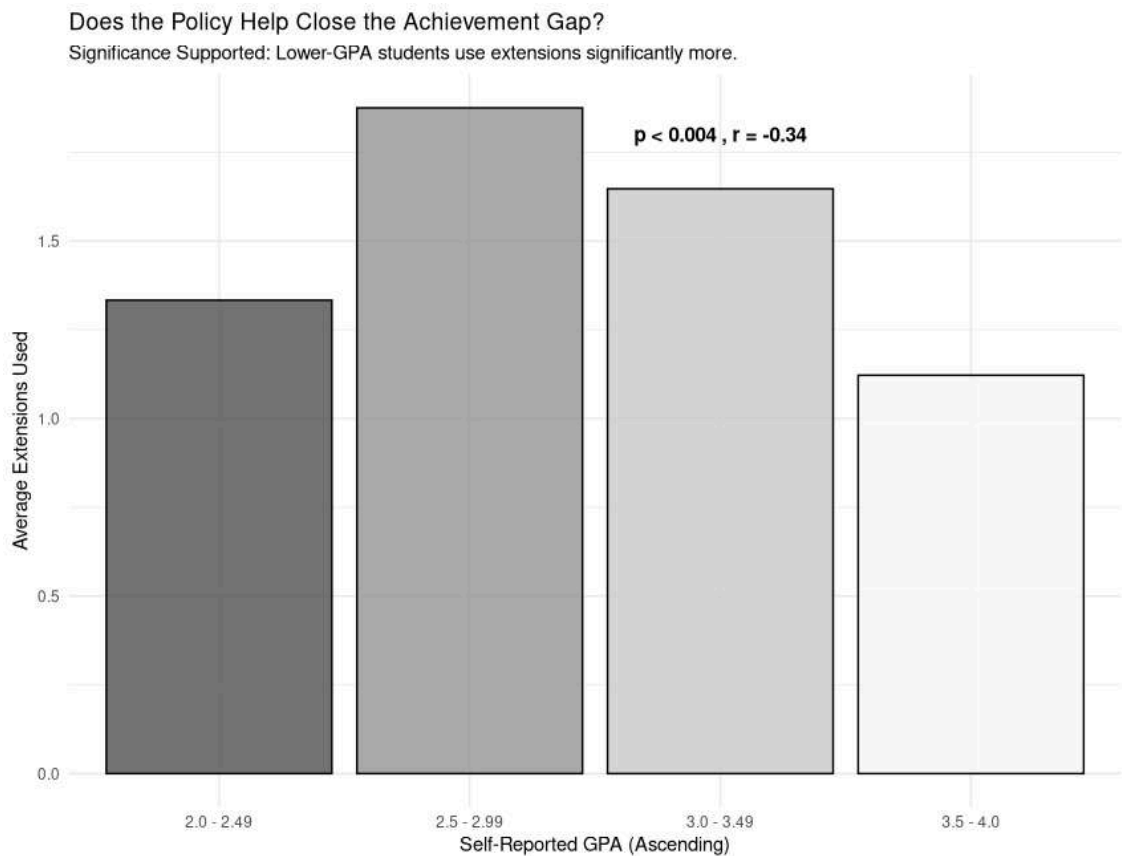


Figure 1: Average extensions used per GPA range.

Graph 1 Description

Bar chart — Average number of extensions used per GPA range.

X-axis: Self-reported GPA (ascending)

Y-axis: Average extensions used

Graph 2 — Final Grade by GPA

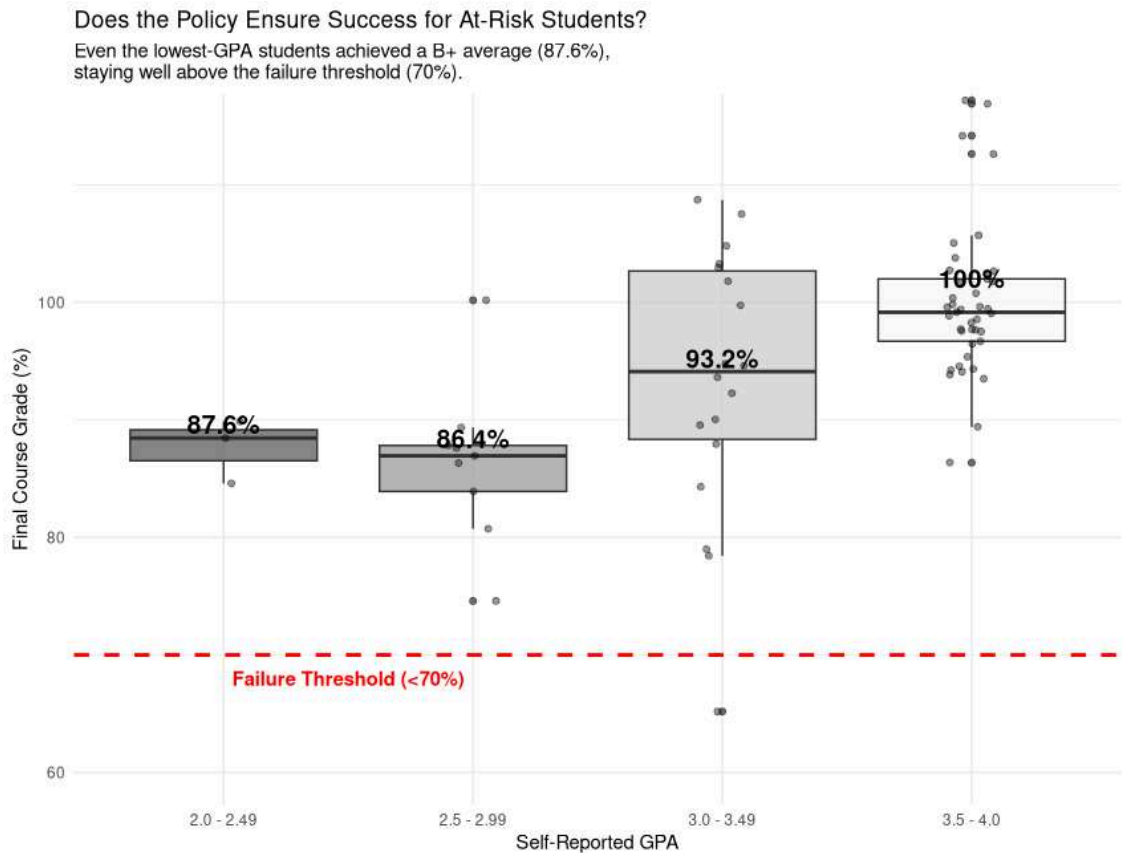


Figure 2: Final course grade (%) per GPA range, with failure threshold.

Graph Description

Boxplot with jitter — Final course grade (%) per GPA range.

X-axis: Self-reported GPA

Y-axis: Final course grade (%)

Mean score labels shown above each box. Red dashed line at $y = 70$ (failure threshold).

Raw Values Summary (for Graph 2)

1. The 2.0 – 2.49 GPA Group (count: 3) had a mean score of 87.6
2. The 2.5 – 2.99 GPA Group (count: 9) had a mean score of 86.9
3. The 3.0 – 3.49 GPA Group (count: 18) had a mean score of 93.2
4. The 3.5 – 4.0 GPA Group (count: 41) had a mean score of 100.0

Statistical Note

Test used: Spearman Correlation (Graph 1)

Result:

$$p < 0.004, r = -0.34$$

Conclusion

A significant correlation is shown: as a student's self-report of their GPA lowers, the higher amount of extensions they will use. Due to our smaller amount of data ($n = 105$), the graph does not seem to show much of a difference. To prove that the extension policy helps students who have a lower self-reported GPA, the second graph shows that the 2 lower GPA groups still got around an 85% as their Final Grade, which is still a good grade in our University.